

Stewart C. Jamieson

PHD IN AUTONOMOUS SYSTEMS · HEAD OF TECHNOLOGY · RISK-AWARE AI & ROBOTICS

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A technical leader and innovator guiding a world-class team of scientists and engineers in developing algorithms that produce safer, more reliable, “uncertainty-aware” artificial intelligences.

Work & Research Experience

Themis AI Inc.

Cambridge, MA, USA

HEAD OF TECHNOLOGY

Apr. 2024 - Present

- Leading the development of Capsa, an on-prem SaaS product platform that helps to eliminate risk from AI systems
- Applying Ph.D. expertise in risk-based machine learning to guide the research and product innovation teams
- Working directly with customers in various sectors to deploy Capsa across a range of diverse use cases

WHOI's Autonomous Robotics and Perception Laboratory (WARPLab)

Woods Hole, MA, USA

GRADUATE RESEARCH ASSISTANT

June 2018 - Mar. 2024

- Developing autonomous exploration algorithms for multi-robot teams exploring the deep ocean and coral reefs
- Publications focus on optimizing robot-human collaboration in challenging communication environments
- Assisting with the deployment of these novel algorithms into WHOI's world-class deep sea exploration vehicles

aUToronto: The University of Toronto's Self-Driving Car Team

Toronto, ON, Canada

SOFTWARE SUB-TEAM LEAD

June 2017 - June 2018

- Led a sub-team of 12 graduate and undergraduate students in a competition to develop an autonomous Chevrolet Bolt
- My team created the overall system software architecture, sensor drivers, vehicle control interface, and software services
- By the end of my term, aUToronto won 1st place in the SAE/GM AutoDrive Challenge (Year 1)

Zebra Technologies Inc.

Mississauga, ON, Canada

SOFTWARE ENGINEERING INTERN, ENGINEERING PRODUCT INNOVATION TEAM

May 2016 - Aug. 2017

- Co-designed and co-developed core software for the first Zebra SmartSight™ robot prototype
- Researched and presented business applications for robotics, machine learning, and neural networks
- 16 months of C++14 development experience including networking, databases, and high performance computing

Wattpad Inc.

Toronto, ON, Canada

ANDROID SOFTWARE DEVELOPER INTERN, READER ACQUISITION TEAM

May 2015 - Sept. 2015

- Implemented features designed to attract new users to Wattpad, a worldwide storytelling platform
- Used A/B testing to validate features before releasing to a community of over 80 million users

QA SOFTWARE DEVELOPER INTERN, ANDROID CORE TEAM

May 2014 - Sept. 2014

- Searched for, reported, and fixed software bugs in the Wattpad Android application which had over 15 million users
- Designed and implemented a virtual doorman to greet company visitors and notify staff of their arrival

Education

Massachusetts Institute of Technology & Woods Hole Oceanographic Institution

Cambridge, MA, USA

PH.D. IN AERONAUTICS AND ASTRONAUTICS — AUTONOMOUS SYSTEMS (5.0/5.0 CGPA)

June 2020 - Mar. 2024

- Thesis: *Enabling Human-Multi-Robot Collaborative Visual Exploration in Underwater Environments*
- Showed risk-based value-of-information enables efficient human-multi-robot interactions in autonomous exploration
- Co-supervised by Prof. Jonathan How (MIT) and Dr. Yogesh Girdhar (Woods Hole Oceanographic Institution)

S.M. IN AERONAUTICS AND ASTRONAUTICS — AUTONOMOUS SYSTEMS (5.0/5.0 CGPA)

June 2018 - May 2020

- Relevant Coursework: Cognitive Robotics, Visual Navigation for Autonomous Vehicles, Bayesian Modelling & Inference
- Co-supervised by Prof. Jonathan How (MIT) and Dr. Yogesh Girdhar (Woods Hole Oceanographic Institution)

University of Toronto

Toronto, ON, Canada

B.A.SC. IN ENGINEERING SCIENCE WITH HONOURS — ROBOTICS MAJOR (3.83/4.0 CGPA)

Sept. 2013 - Apr. 2018

- Thesis: *Deep Learning for Robust Vision in Realtime Autonomous Driving*, supervised by Prof. Angela Schoellig
- Applied state-of-the-art uncertainty quantification techniques to achieve safer autonomous driving performance

Research Interests & Skills

Artificial Intelligence	Risk-Aware Online Learning, Bayesian Modelling & Inference, Reinforcement Learning, AI Ethics
Robotics	Human-Robot Collaboration, 3-D Semantic Mapping, Robust Vision, Informative Path Planning
Programming	Python 3, C++20, PyTorch, ROS, OpenCV, MATLAB, Mathematica, Android/Java

Publications

JOURNAL PAPERS

Jamieson, S., How, J. P., & Girdhar, Y. (2024). *Finding the optimal exploration-exploitation trade-off online through Bayesian risk estimation and minimization*. In Artificial Intelligence, vol. 330, May 2024. DOI: 10.1016/j.artint.2024.104096.

PEER-REVIEWED CONFERENCE PAPERS

McCammon, S., **Jamieson, S.**, Mooney, T. A., & Girdhar, Y. (2024). *Discovering Biological Hotspots with a Passively Listening AUV*. In 2024 IEEE International Conference on Robotics and Automation (ICRA). Yokohama, Japan.

Jamieson, S., How, J. P., & Girdhar, Y. (2023). *DeepSeeColor: Realtime Adaptive Color Correction for Autonomous Underwater Vehicles via Deep Learning Methods*. In 2023 IEEE International Conference on Robotics and Automation (ICRA). London, UK.

Girdhar, Y., McGuire, N., Cai, L., **Jamieson, S.**, McCammon, S., San Soucie, J. E., Todd, J. E., Claus, B., & Mooney, T. A. (2023). *CUREE: A Curious Underwater Robot for Ecosystem Exploration*. In 2023 IEEE International Conference on Robotics and Automation (ICRA). London, UK.

Jamieson, S., Fathian, K., Khosoussi, K., How, J. P., & Girdhar, Y. (2021). *Multi-Robot Distributed Semantic Mapping in Unfamiliar Environments through Online Matching of Learned Representations*. In 2021 IEEE International Conference on Robotics and Automation (ICRA). Xi'an, China.

Jamieson, S., How, J. P., & Girdhar, Y. (2020). *Active Reward Learning for Co-Robotic Vision Based Exploration in Bandwidth Limited Environments*. In 2020 IEEE International Conference on Robotics and Automation (ICRA). Paris, France. **Won Best Paper Award in Service Robotics.**

Girdhar, Y., Cai, L., **Jamieson, S.**, McGuire, N., Flaspohler, G., Suman, S., & Claus, B. (2019). *Enabling Co-Robotic Scientific Exploration of Unknown Environments over a Low Bandwidth Communication Channel*. In 2019 IEEE International Conference on Robotics and Automation (ICRA). Montréal, Canada.

WORKSHOP PAPERS

Yang, D., Cai, L., **Jamieson, S.**, & Girdhar, Y. (2023). *Robot Goes Fishing: Rapid, High-Resolution Biological Hotspot Mapping in Coral Reefs with Vision-Guided Autonomous Underwater Vehicles*. In "CV4Animals" Workshop at CVPR 2023. Vancouver, Canada.

Jamieson, S., Todd, J. E., How, J. P., & Girdhar, Y. (2021). *Communicating Efficiently to Enable Human-Multi-Robot Collaboration in Space Exploration*. In "SpaceCHI: Human-Computer Interaction for Space Exploration" Workshop at CHI 2021. Yokohama, Japan.

Beaulieu, S., Alexander, H., **Jamieson, S.**, Longworth, B., McLean, C., Soenen, K., York, A., Krinos, A., Cai, L., Govostes, R., & Hernandez, C. (2020). *Building a data science curriculum and community for ocean scientists, engineers, and students using The Carpentries model*. In AGU Fall Meeting 2020.

Jamieson, S. (2019). *The Pervasiveness of Deep Learning in Robotics Research Does Not Impede Scientific Insights into Robotics Problems*. In "Debates on the Future of Robotics Research" Workshop at ICRA 2019. Montréal, Canada.

THESES

Jamieson, S. (2024). *Enabling Human-Multi-Robot Collaborative Visual Exploration in Underwater Environments*. Doctoral Thesis, Massachusetts Institute of Technology & Woods Hole Oceanographic Institution.

Jamieson, S. (2020). *Enabling Human-Robot Cooperation in Scientific Exploration of Bandwidth-Limited Environments*. Master's Thesis, Massachusetts Institute of Technology & Woods Hole Oceanographic Institution.

Jamieson, S. (2018). *Deep Learning for Robust Vision in Realtime Autonomous Driving*. B.A.Sc. Thesis, University of Toronto.

Honors & Awards

INTERNATIONAL

- 2020 **Best Paper Award in Service Robotics (out of 1483 Accepted Papers)**, ICRA 2020 [Paris, France](#)
2018 **1st Place Team**, SAE/GM AutoDrive Challenge [Yuma, AZ, USA](#)

ACADEMIC

- 2013 **Governor General's Bronze Medal for Academic Excellence**, Corpus Christi Secondary School [Burlington, Canada](#)
2013 **Regional Champion**, ECOO Programming Competition [Halton Region, Canada](#)
2010-13 **School Champion (x4)**, Waterloo CEMC Math Contest [Burlington, Canada](#)

Presentations

Videos and other materials used in some of the following presentations are available at www.stewartjamieson.com

INVITED AND SELECTIVE PRESENTATIONS

- T-Challenge Competition (T-Mobile & Deutsche Telekom)** [Bonn, Germany](#)
AWARD WINNER AND PRESENTER — *Themis AI's Capsa: Making AI Trustworthy* Jun. 2025
- CSAIL Alliances Annual Meeting** [Cambridge, MA, USA](#)
SELECTED STARTUP PITCH PRESENTER Apr. 2025
- MIT Digital Technology and Strategy Conference** [Cambridge, MA, USA](#)
SELECTED LIGHTNING TALK PRESENTER Sept. 2024
- NCCOS SfM/AI Working Group** [Online](#)
INVITED TALK — *Multi-Robot Reef Monitoring, Mapping, & Human-Guided Exploration* Mar. 2023
- 15th International Coral Reef Symposium** [Bremen, Germany](#)
INVITED TALK — *Rapid, Comprehensive Coral Reef Monitoring with AUVs* Jul. 2022
- WHOI AOPE Department Seminar Series** [Woods Hole, MA, USA](#)
INVITED TALK — *Human-Robot Cooperation for Exploring Bandwidth-Limited Environments* Jul. 2020
- ICRaxMIT** [Cambridge, MA, USA](#)
INVITED PRESENTER & PANELIST — *Active Reward Learning for Co-Robotic Exploration* Jun. 2020
- Debates on the Future of Robotics Research, ICRA** [Montréal, QC, Canada](#)
INVITED LIGHTNING TALK — *Deep Learning in Robotics Research* May 2019

INDUSTRY AND PUBLIC OUTREACH PRESENTATIONS

- CXO Forum 2025 (CCHL & Digital Health Canada)** [Halifax, NS, Canada](#)
INVITED SPEAKER — *AI Trust, Privacy, & Where Do We Go From Here?* Oct. 2025
- Ai4 Conference** [Las Vegas, NV, USA](#)
SPEAKER (INDUSTRY PRESENTATION) — *Capsa: Making AI Trustworthy* Aug. 2024
- WHOI Ocean Encounters (Season 5, Episode 4)** [Online](#)
INVITED PRESENTER — *ROBOTS: How New Technologies Are Advancing Ocean Research* May 2023
- Massachusetts Institute of Technology** [Cambridge, MA, USA](#)
CO-LECTURER (COURSE PRESENTATION) — *Multi-Robot Adaptive Sampling* Apr. 2019
- Zebra Technologies Inc. "Lunch & Learn" (200+ attendees)** [Mississauga, ON, Canada](#)
LEAD PRESENTER (INTERNAL SEMINAR) — *Introduction to Neural Networks and Machine Learning* Jan. 2017

POSTERS AND OTHER PRESENTATIONS

- SpaceCHI Workshop @ CHI** [Online](#)
POSTER PRESENTATION — *Communication for Human-Multi-Robot Collaboration* May 2021
- University of Toronto Debate** [Toronto, ON, Canada](#)
CO-PRESENTER — *Should Robots Have Rights?* Dec. 2015
- Praxis II Showcase, University of Toronto** [Toronto, ON, Canada](#)
CO-PRESENTER — *A Customized Graphical Checklist for Efficient Ambulance Inventory* Apr. 2014

Academic Service

- Co-Organizer of:
 - Robots for Understanding Natural Ecosystems Workshop @ ICRA 2024
 - AI for Earth & Space Science Workshop @ ICLR 2022
- Journal Reviewer:
 - Journal of Aerospace Information Systems (JAIS)
 - Robotics and Automation Letters (RA-L)
 - IEEE Transactions on Robotics (T-RO)
- Conference Reviewer:
 - Conference on Control Applications in Marine Systems (CAMS)
 - International Conference on Learning Representations (ICLR)
 - International Conference on Machine Learning (ICML)
 - International Conference on Robotics and Automation (ICRA)
 - International Conference on Intelligent Robots and Systems (IROS)
 - Conference on Neural Information Processing Systems (NeurIPS)

Teaching

The Carpentries

CERTIFIED SOFTWARE CARPENTRIES INSTRUCTOR

- Co-Instructor for the WHOI Student Python Workshop Series 2021
- Helper for the WHOI Data Carpentry Workshop Fall 2020
- Co-Instructor for the WHOI Student Python Workshop Series 2020
- Helper for the WHOI Software Carpentry Workshop Fall 2019

Massachusetts Institute of Technology

TEACHING ASSISTANT

- 16.485 Visual Navigation for Autonomous Vehicles (Lead Instructor: Prof. Luca Carlone)

Worldwide

Oct. 2019 - Present

Cambridge, MA, USA

Aug. 2020 - Dec. 2020

Professional Service

MIT-WHOI Joint Program

ELECTED AT-LARGE REPRESENTATIVE

- Co-hosted monthly online student events during the COVID-19 pandemic
- Developed and published the MIT-WHOI Joint Program online photoboard

Zebra Technologies Inc.

EDITOR, EMC INNOVATION NEWSLETTER

- Edited bi-monthly department newsletter and distributed it to over 1700 engineers
- Commissioned, reviewed, and published articles about recent trends and innovations in electronics, robotics, etc.

IEEE (Institute of Electrical and Electronics Engineers)

MEMBER

Woods Hole, MA, USA

Oct. 2020 - Oct. 2021

Mississauga, ON, Canada

May 2016 - Aug. 2017

Worldwide

Nov. 2013 - Present